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05/08/2026

Optimizing Customer Support Through Generative AI

A Data-Driven Framework for AI Workflow Prioritization

BUSINESS PROBLEM

Which customer support workflows should be Automated, AI-Assisted, or Human-Led?

Customer Support Pressure

- Rising ticket volume
- Higher customer expectations
- Increasing operational costs

Generic AI Deployment

- Same AI for every workflow
- One-size-fits-all automation

Business Risk

- Misallocated AI investment
- Poor customer experience
- Increased operational risk

PROJECT APPROACH

Develop a data-driven framework for prioritizing Generative AI across customer support workflows.

Dataset

- 85K customer support interactions

Data Exploration

- Understand data quality and structure

Hypothesis Testing

- Can customer remarks support NLP?

Operational Analysis

- Identify high-impact workflows

AI Prioritization Framework

- Classify workflows by automation potential

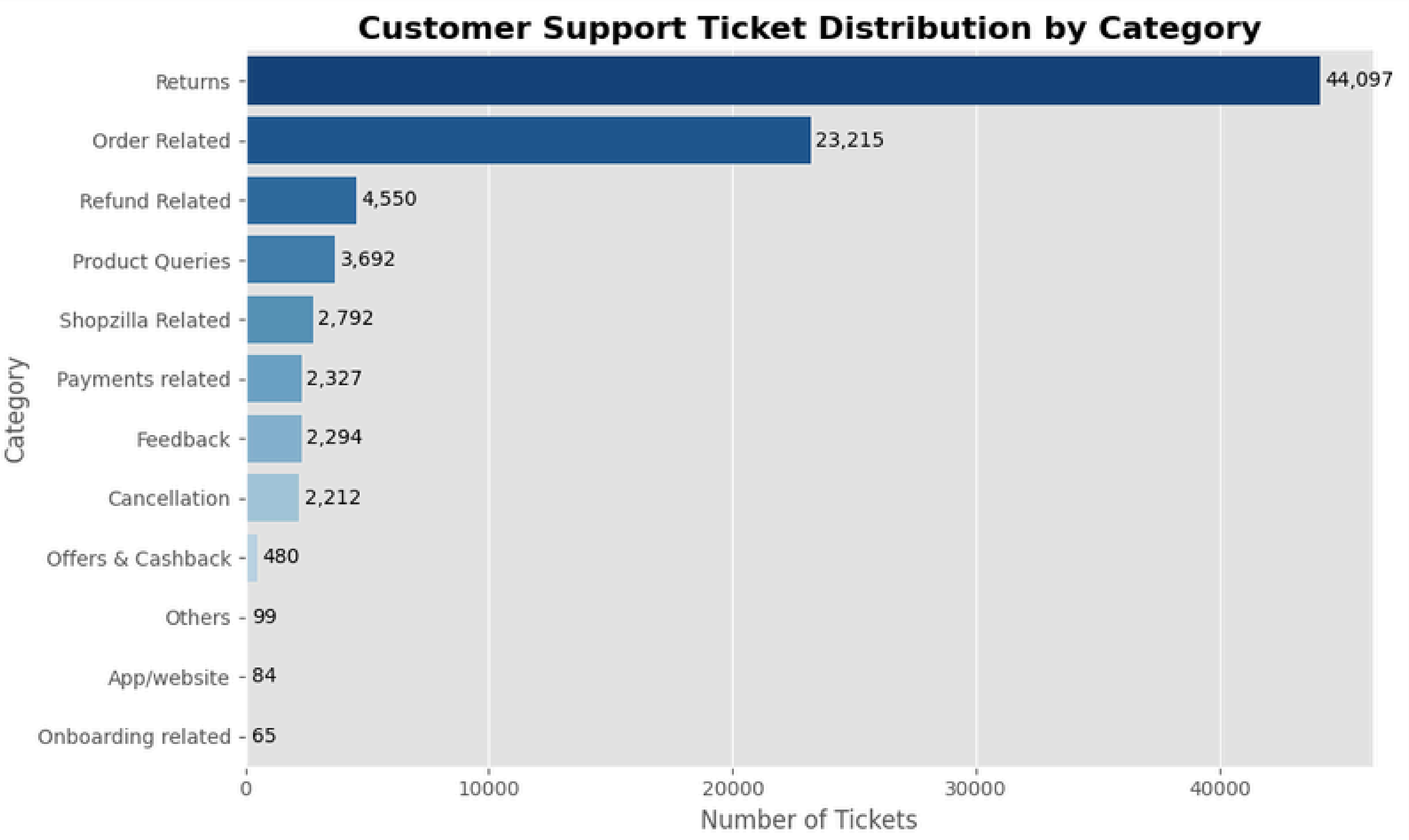
Machine Learning Validation

- Validate findings using LightGBM + SHAP

Explainable AI

- Translate machine learning predictions into actionable business insights.

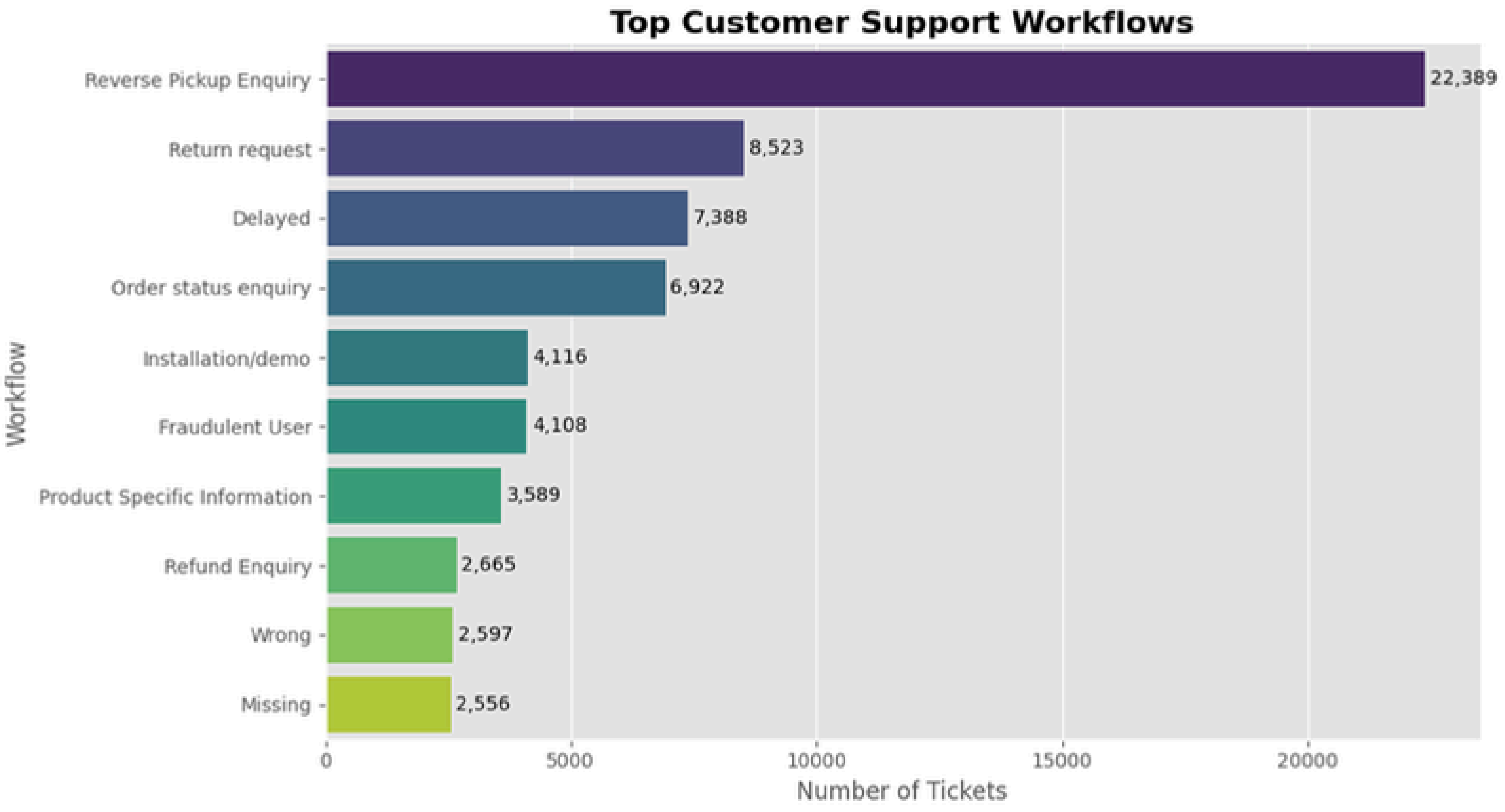
Understanding the Customer Support Landscape



INSIGHTS

- Customer support demand is highly concentrated, with Returns and Order Related accounting for approximately 84% of all support tickets.
- Rather than optimizing every support process equally, focusing on the highest-volume categories offers the greatest opportunity for operational impact.
- These findings established the foundation for identifying which specific workflows should be prioritized for Generative AI implementation.

Workflow Demand

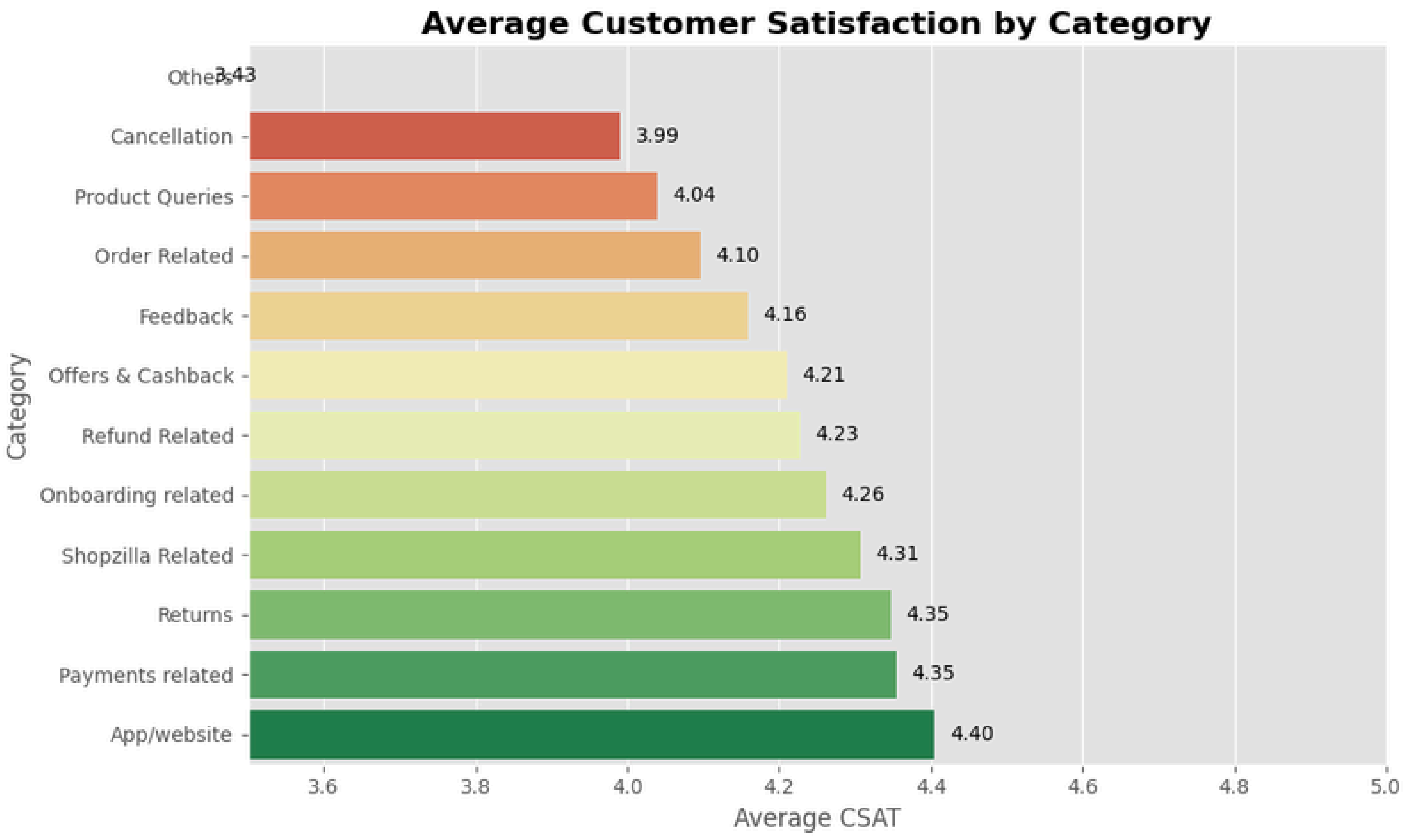


INSIGHTS

- Reverse Pickup Enquiry is the largest workflow, with over 22K tickets, making it the biggest operational workload.
- The top workflows are mostly post-purchase issues: pickup, returns, delays, order status, installation, and refunds.
- This suggests GenAI should focus on operational support workflows, not just generic chatbot responses.

AI value is highest where repetitive workflows consume the most support effort.

Customer Satisfaction Across Support Categories



INSIGHTS

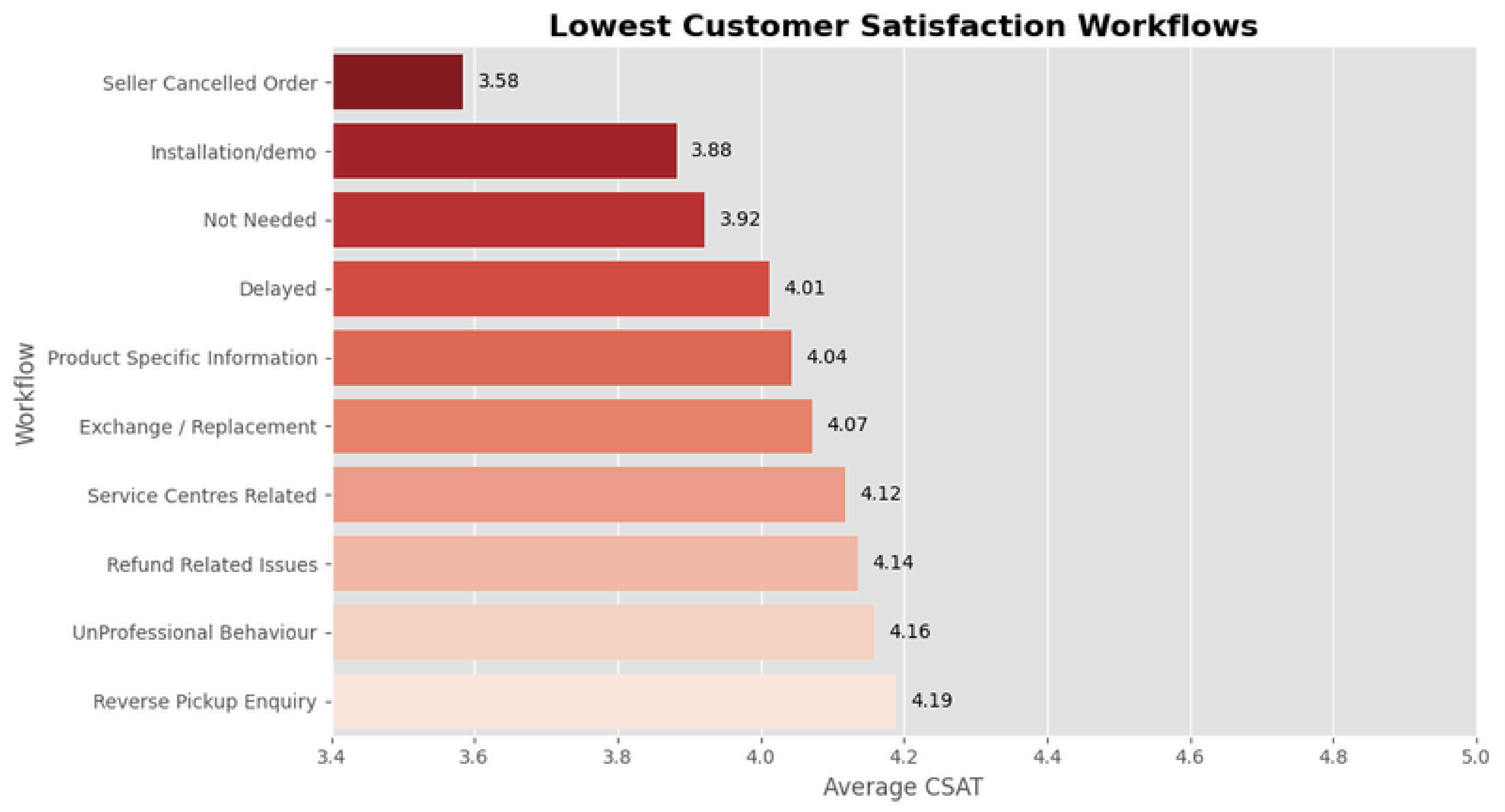
Customer satisfaction is not uniform across customer support categories, indicating that different workflows create different customer experiences.

Categories related to post-purchase operations, such as returns and order-related issues, exhibit comparatively lower satisfaction, highlighting opportunities for process improvement.

These differences suggest that AI should be deployed selectively, prioritizing workflows where operational friction has the greatest impact on customer experience.

Customer satisfaction varies by workflow, reinforcing the need for a targeted rather than one-size-fits-all AI strategy.

Lowest Customer Satisfaction Workflows



INSIGHTS

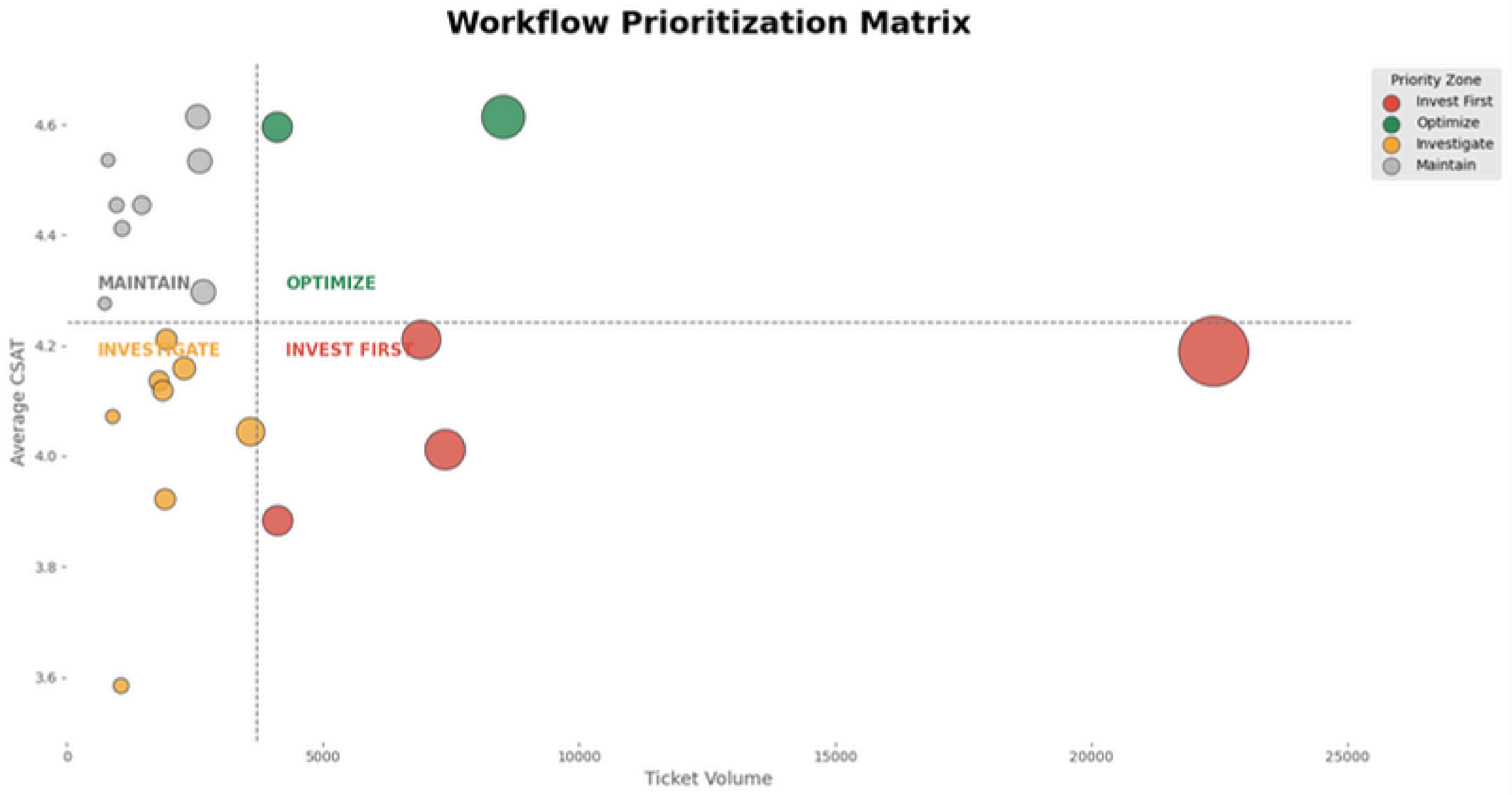
Customer dissatisfaction is concentrated within a small number of operational workflows, rather than being evenly distributed across customer support.

Workflows such as Delayed Orders, Installation & Demo, Seller Cancelled Orders, and Refund-related requestsconsistently exhibit lower customer satisfaction, indicating higher operational friction.

These workflows represent the highest-priority opportunities for AI-assisted process improvement, where reducing delays and improving operational coordination can have the greatest impact.

Targeting low-CSAT operational workflows offers the greatest opportunity to improve customer experience through Generative AI.

Prioritizing Workflows for AI Investment

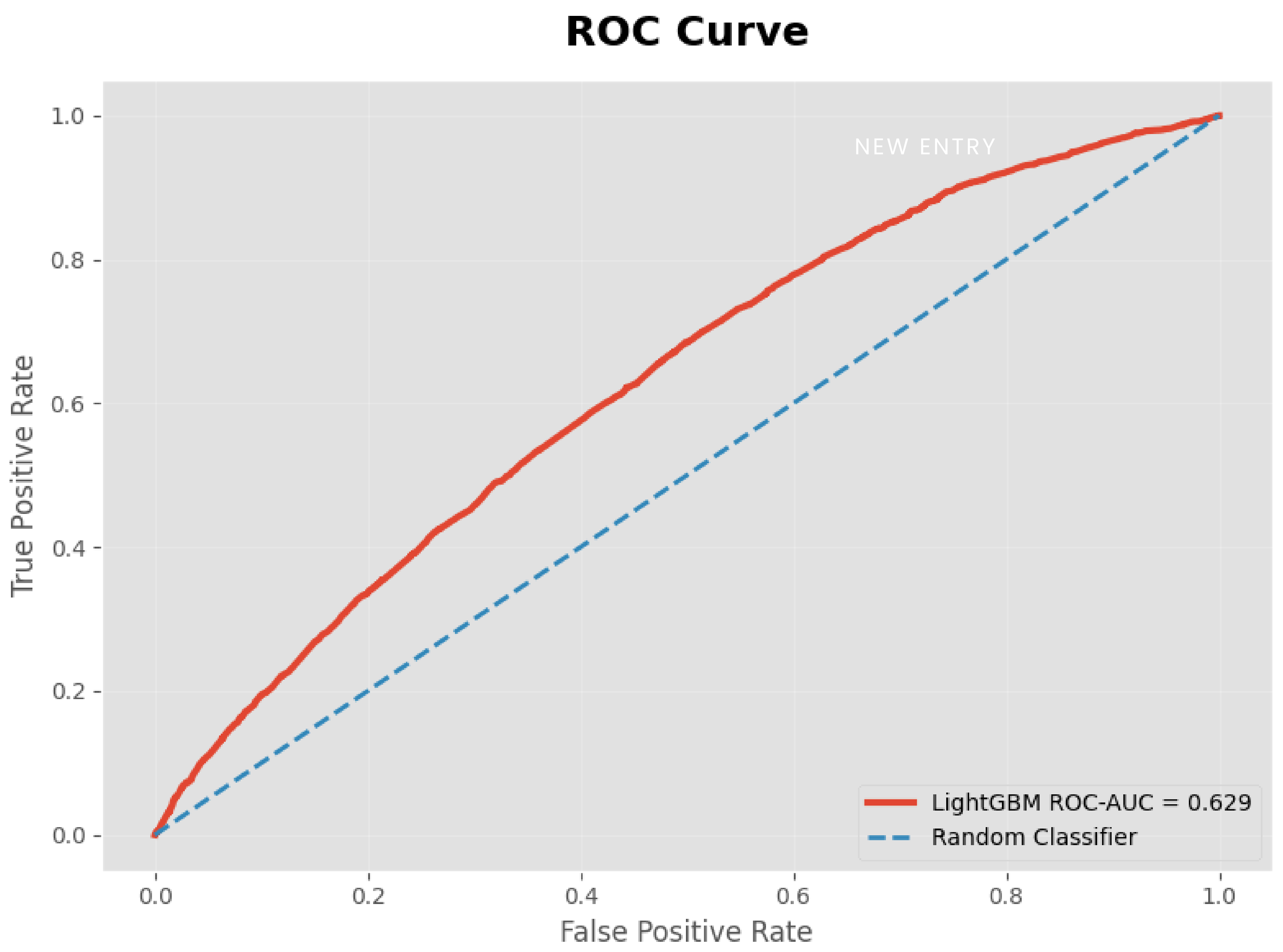


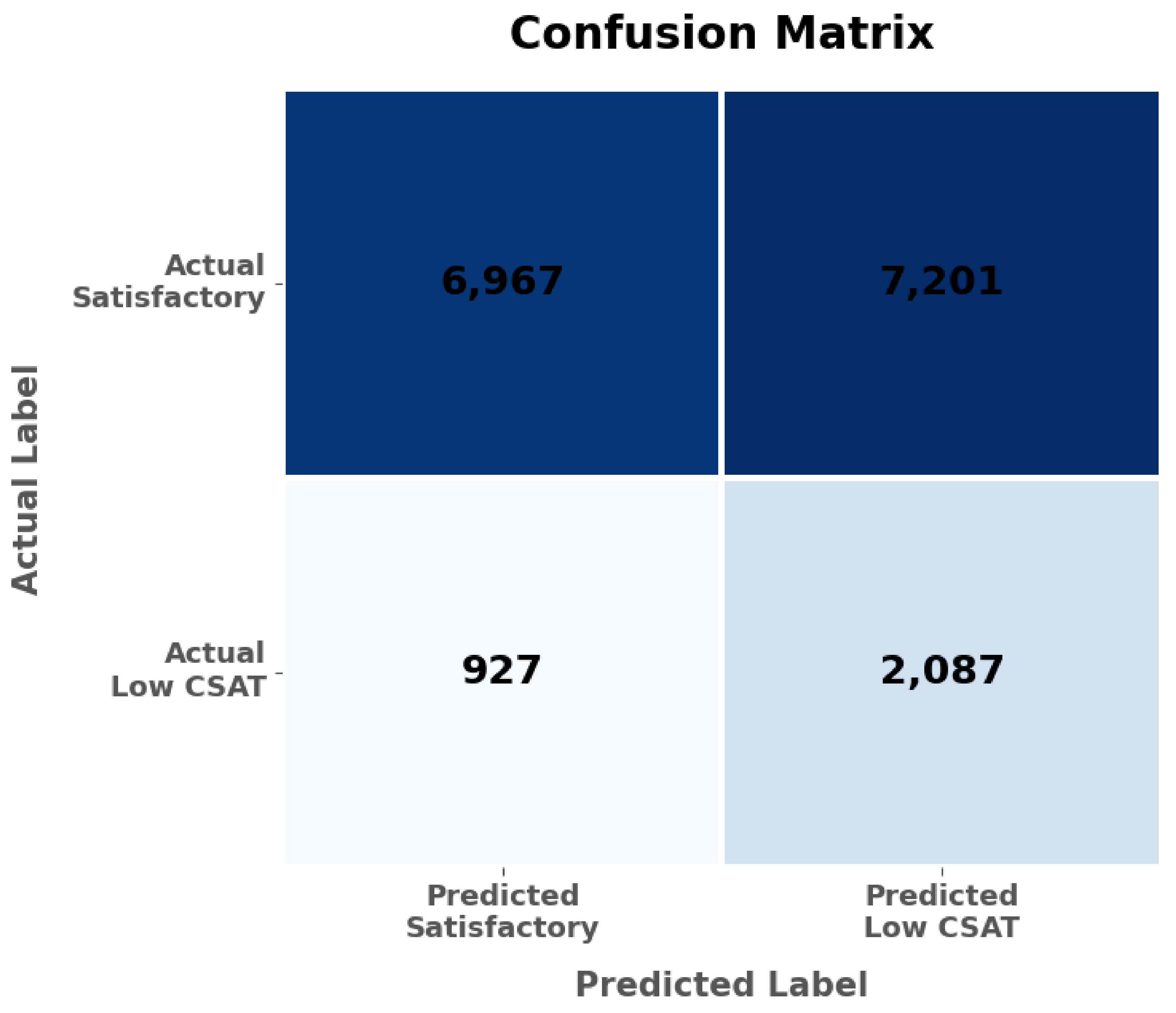
INSIGHTS

- AI priority depends on both ticket volume and customer satisfaction, not either metric alone.
- Delayed Orders emerged as a key AI-assisted opportunity due to high volume and weaker CSAT.
- Lower-volume but poor-CSAT workflows should be investigated for targeted process improvement.

High-volume, low-CSAT workflows are the strongest candidates for AI-assisted support.

Validating the Framework with Machine Learning





- A LightGBM classifier was trained to identify tickets at risk of low satisfaction.
- The model achieved ROC-AUC = 0.629, showing that operational features contain meaningful predictive signal.
- The baseline model slightly outperformed the tuned model, so the simpler model was retained.

The model confirmed that customer satisfaction is linked to operational workflow patterns, not random ticket variation.

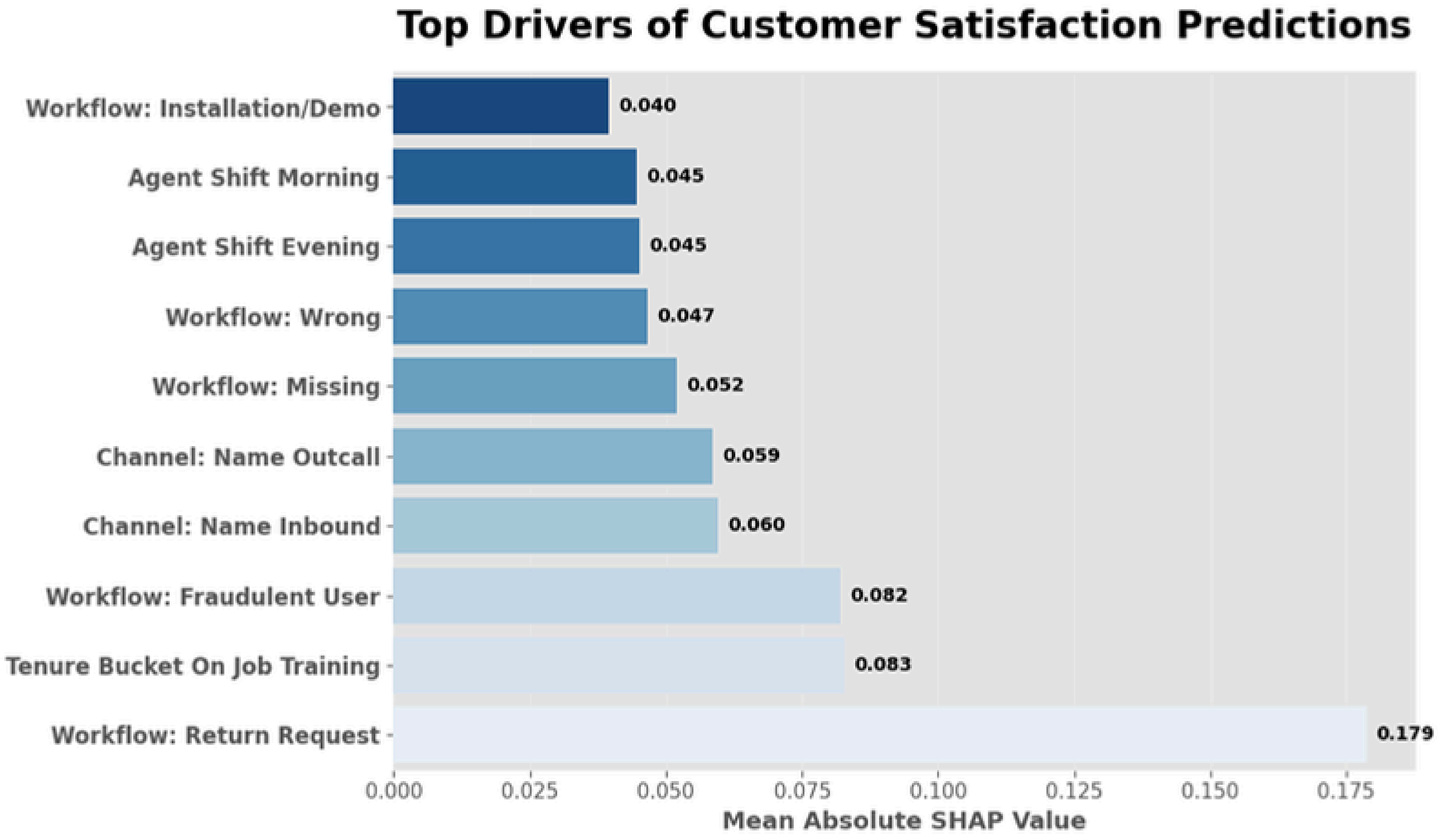
What Drives Customer Satisfaction?

Explainable AI using SHAP

- SHAP analysis identified the operational factors that contributed most strongly to customer satisfaction predictions.
- Workflow type and employee tenure emerged as stronger predictors than generic ticket characteristics.
- These findings reinforce that operational process design, rather than customer behavior alone, drives support outcomes.

Explainable AI transformed the model from a predictive tool into an actionable decision-support framework by identifying the operational drivers of customer satisfaction.

Which Factors Influenced the Model Most?



- Workflow-level variables were among the strongest drivers of customer satisfaction predictions.
- This confirms that customer experience is shaped by operational process design, not only by generic ticket attributes.
- SHAP helped translate the model output into interpretable business drivers for AI prioritization.

The model validated that operational workflows are meaningful predictors of customer satisfaction.

Recommended AI Operating Model

The goal is not to automate every ticket, but to match AI involvement to workflow complexity and risk.

Fully Automate	AI-Assisted	Human-Led
Order Status Enquiries	Delayed Orders	Fraud Investigations
Invoice Requests	Installation / Demo	Seller Disputes
Product Information	Refund Processing	Policy Exceptions
Basic Tracking Requests	Reverse Pickup Enquiries	Complex Escalations

Project Outcomes & Business Impact

Key Outcomes		Expected Business Impact
Identified the highest-volume customer support workflows	01	Faster response times through intelligent automation
Prioritized AI opportunities using ticket volume and customer satisfaction	02	Reduced workload for customer support agents
Validated findings using a LightGBM machine learning model	03	Improved customer satisfaction in high-friction workflows
Used SHAP explainability to identify the key operational drivers of customer satisfaction	04	More consistent and scalable customer support operations
Developed a practical framework for prioritizing customer support workflows based on business impact and AI suitability.	05	Practical roadmap for responsible Generative AI adoption